

Fermi–Amaldi-Based Local Mixing Function

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(Dated: March 12, 2026)

I. MAIN TEXT

Consider the exact exchange energy density (closed-shell):

$$e_x^{\text{exact}}(\mathbf{r}) = -\frac{1}{4} \int \frac{|\gamma(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}', \quad (1)$$

where

$$\gamma(\mathbf{r}, \mathbf{r}') = 2 \sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r}) \varphi_i(\mathbf{r}') \quad (2)$$

is one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N ,

$$e_x^{\text{FA}}(\mathbf{r}) = -\frac{1}{2N} \int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}'. \quad (3)$$

For one-orbital systems (one-electron and two-electron singlet), two quantities are equal for all values of $\mathbf{r}$ [5–7]:

$$e_x^{\text{exact}}(\mathbf{r}) = e_x^{\text{FA}}(\mathbf{r}) \quad N \leq 2 \quad \text{for all } \mathbf{r}. \quad (4)$$

For systems with more than one molecular orbital, the spatial behavior of both quantities needs to be studied closer to establish a relation similar to Eq. (4). Which is convenient to do analysing the corresponding exchange holes. The exchange hole is defined via

$$e_x(\mathbf{r}) = \frac{1}{2} \int \frac{\rho(\mathbf{r})\rho_x(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \quad (5)$$

leading to expressions for the exact and the Fermi–

Amaldi exchange holes, respectively:

$$\rho_x^{\text{exact}}(\mathbf{r}, \mathbf{r}') = -\frac{1}{2} \frac{|\gamma(\mathbf{r}, \mathbf{r}')|^2}{\rho(\mathbf{r})}, \quad (6)$$

$$\rho_x^{\text{FA}}(\mathbf{r}') = -\frac{1}{N} \rho(\mathbf{r}'). \quad (7)$$

Both quantities are strictly non-positive, hence, we will consider their non-negative absolute values, $|\rho_x^{\text{exact}}(\mathbf{r})|$ and $|\rho_x^{\text{FA}}(\mathbf{r})|$, for convenience.

Effectively, comparing equations (6) and (7), we are looking for zeros of the equation:

$$N|\gamma(\mathbf{r}, \mathbf{r}')|^2 - 2\rho(\mathbf{r})\rho(\mathbf{r}') = 0 \quad N > 2, \quad (8)$$

where, by Cauchy–Schwarz inequality:

$$\rho(\mathbf{r})\rho(\mathbf{r}') \geq |\gamma(\mathbf{r}, \mathbf{r}')|^2 \quad N > 2 \quad \text{for all } \mathbf{r}. \quad (9)$$

In the limit of infinitesimally small interelectronic distances:

$$\lim_{|\mathbf{r}-\mathbf{r}'| \rightarrow 0} \gamma(\mathbf{r}, \mathbf{r}')\gamma(\mathbf{r}', \mathbf{r}) = \rho(\mathbf{r})\rho(\mathbf{r}'), \quad (10)$$

thus, the well of the exchange hole is dominated by the exact exchange hole.

II. SCRAP

A local hybrid exchange functional is given by

$$E_x^{\text{LH}} = \int \{g(\mathbf{r})e_x^{\text{exact}}(\mathbf{r}) + [1 - g(\mathbf{r})]e_x^{\text{SL}}(\mathbf{r})\} d\mathbf{r}, \quad (11)$$

where the integrand is a mixture of two exchange density energies weighed by a local mixing function (LMF), $g(\mathbf{r})$, which is position-dependent. The e_x^{SL} and e_x^{exact} quantities are a semi-local and the exact exchange energy densities, respectively.

A particular case of a one-electron limit, i.e., the asymptotic limit ($r \rightarrow \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r \rightarrow \infty} \frac{e_x^{\text{FA}}}{e_x^{\text{exact}}} = 1 \quad (12)$$

since both energy densities approach $-1/2r$ as $r \rightarrow \infty$.

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